

No universal estimator of $n:m$ phase coupling: complementary metric blind spots and a validated cross-signal detector

Antoine Bellemare-Pepin

collaborators

Draft for internal review.

Many biological rhythms do not merely co-occur but lock into integer $n:m$ polyrhythms: the heart beats a fixed number of times per breath, cortical activity phase-locks to tremor, and stride entrains to ventilation. The coupling *order* and its stability are clinically meaningful state variables — they grade anaesthetic depth, index autonomic and sleep regulation, track fetal maturation, and scale with tremor severity. Yet these estimators are rarely validated against ground truth, and they face a fundamental confound: a single non-sinusoidal oscillation decomposes into harmonics that are intrinsically phase-locked, so apparent $n:m$ coupling can arise with no second oscillator at all. We treat $n:m$ coupling *measurement* as an object of study, on synthetic ground truth and real recordings, and reach two organizing results. First, **no single estimator is universal**: the $n:m$ phase-locking value (PLV) is the most sensitive detector on clean, unimodal locks and at low signal-to-noise, but is blind by construction to *multimodal* locks, where entropy- and mutual-information indices recover the coupling it misses — so sound measurement needs a metric *panel*, not a single statistic. Second, **the harmonic confound is a matter of scope**: within one non-sinusoidal signal every technique false-positives and no standard surrogate fully removes it, so $n:m$ coupling is soundly interpretable only between independent sources. Along the way we isolate the estimation choices that separate recovering a clean polyrhythm from missing it entirely — the ratio convention, the phase estimator, ratio coverage, and readout weighting — and give a corrected recipe that matches an oracle on ground truth. We package the result as a cross-signal detector (a technique panel with surrogate inference and a scope guard) and demonstrate it on real data: genuine cardiorespiratory polyrhythms are recovered; in sleep EEG a within-channel, state-dependent $n:m$ phase-coupling spectrum is recovered and replicated across montages, while inter-regional coupling is correctly returned at the noise floor.

1 Introduction

When two oscillations of different frequency interact, they need not lock only at 1:1. They can settle into a rational $n:m$ regime — n cycles of one rhythm to m of the other — a polyrhythmic form of synchronization. This was first made measurable in physiology by the cardiorespiratory synchrogram, which revealed long, hidden epochs of 2:1 through 5:1 heartbeat-to-breath locking in resting humans (Schäfer et al., 1998, 1999). The same $n:m$ formalism, equipped with a noise-tolerant detector (Tass et al., 1998), exposed pathological cortico-muscular locking in Parkinsonian tremor; $n:m$ coupling has since been documented between maternal and fetal heartbeats (van Leeuwen et al.,

2009), between stride and breath in running (Bramble & Carrier, 1983), and, more contentiously, across frequency bands in the brain (Palva et al., 2005; Belluscio et al., 2012).

The coupling order and its stability are not incidental. The cardiorespiratory ratio shifts in an orderly cascade ($2:1 \rightarrow 3:1 \rightarrow 4:1 \rightarrow 5:1$) with anaesthetic depth; cardiorespiratory phase synchronization indexes vagal/autonomic regulation and is altered in heart failure and diabetic autonomic neuropathy; it is sleep-stage dependent and disrupted in sleep apnoea; maternal-fetal coupling tracks fetal autonomic maturation; and cortico-muscular $n:m$ synchronization scales with tremor severity and is a target for deep-brain stimulation (Schnitzler & Gross, 2005; Purdon et al., 2013). Measuring $n:m$ coupling reliably is therefore of direct clinical and scientific value.

Two problems recur. First, the estimators — the $n:m$ phase-locking value, the Shannon entropy of the phase-difference distribution, conditional-probability indices, and phase mutual information — are assembled from a phase estimator, a ratio rule and a coupling statistic, each with defaults, and applied to data without checking that they recover a *known* polyrhythm. Second, a single non-sinusoidal rhythm contains harmonics that are phase-locked by construction, so any $n:m$ statistic computed within one signal can report “coupling” that is merely waveform shape — the confound emphasized by Aru et al. (2015), Scheffer-Teixeira & Tort (2016) and Cole & Voytek (2017), and compounded by surrogate nulls whose construction is neither standardized nor robust.

Here we make the measurement itself the experiment. Using cardiorespiratory recordings and polyrhythmic ground truth with controllable ratio, coupling strength, waveform and noise, we ask four questions of the $n:m$ technique family: are polyrhythms present and recoverable in real biosignals; which estimation choices decide whether a known lock is retrieved; which technique wins in which regime; and does coupling survive the harmonics confound? The answers, in order: polyrhythms are real and recoverable; four choices — convention, phase estimator, ratio coverage, weighting — are decisive; no single technique is universal; and $n:m$ coupling is trustworthy only between independent sources. We turn each answer into a concrete recommendation and ship a validated cross-signal detector.

2 Results

2.1 Polyrhythms are present in real biosignals

We first verify that the target phenomenon exists and is recoverable. On the PhysioNet *Fantasia* young-adult recordings we extracted the cardiac phase (linear between successive R-peaks) and the respiratory phase (Hilbert analytic phase of cleaned respiration), and tested, per subject, the single $n:m$ ratio predicted by that subject’s heart/breath frequency ratio (avoiding multiple-comparison inflation). A sliding-window synchronization index recovers genuine, intermittent locking: the clearest subject shows a 1:3 lock (three heartbeats per breath, matching the 3.2 frequency ratio) with a windowed resultant reaching $R_{\max} = 0.96$, significant against a cyclic-shift surrogate null ($p = 0.005$; Figure 1). Consistent with the classical finding that cardiorespiratory synchronization is intermittent and subject-dependent (Schäfer et al., 1998, 1999), not every resting subject locks at the predicted ratio. The phenomenon is real; the rest of the paper asks whether our instruments measure it soundly.

2.2 A genuine, state-dependent $n:m$ coupling spectrum in sleep EEG

$n:m$ coupling also appears in the brain, where the slow oscillations of non-REM sleep impose strong low-frequency rhythms. We computed the whole-spectrum $n:m$ phase-coupling spectrum (every frequency pair, fraction-kernel ratios, canonical convention, Hilbert phase, power-independent) per

Polyrhythms are present in biosignals: $n:m$ cardiorespiratory phase coupling (Fantasia)

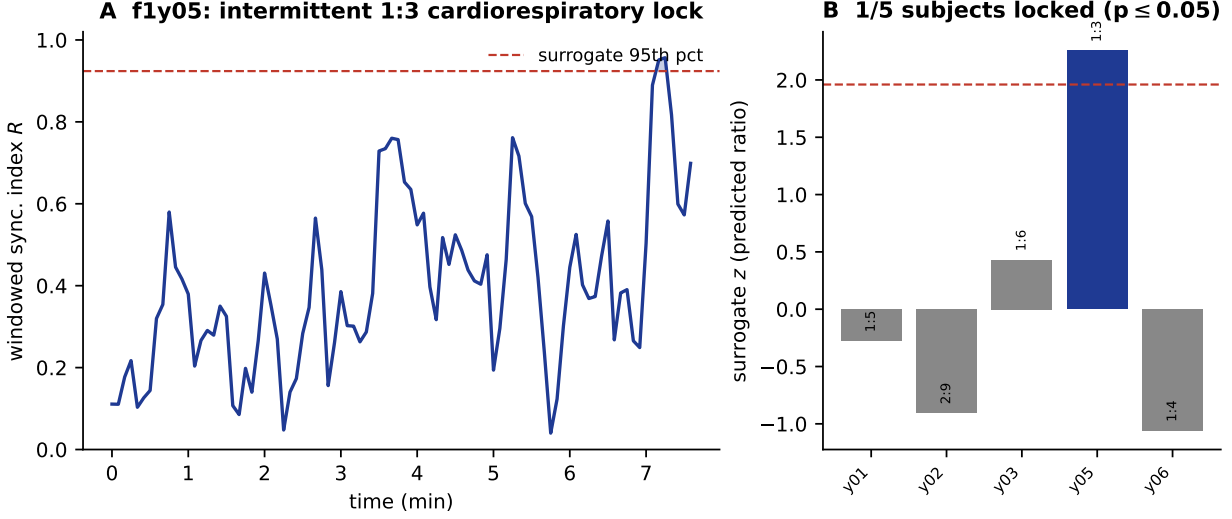


Figure 1: A genuine $n:m$ polyrhythm in a real biosignal. (A) Sliding-window cardiorespiratory synchronization index for the clearest subject: epochs in which the windowed resultant of the 1:3 relative phase exceeds the surrogate 95th percentile mark genuine intermittent locking. (B) Per-subject surrogate z at the frequency-predicted ratio; cardiorespiratory locking is intermittent and subject-dependent.

30 s epoch of the frontal derivation (Fpz–Cz) across Wake/N2/N3/REM (Sleep-EDF), normalising each epoch by a *waveform-preserving* time-shuffle surrogate. Within-channel coupling is strongly stage-dependent in the delta band — 6–7 of 10 frequency bins differ across stages (Kruskal–Wallis, FDR-corrected), strongest near 3 Hz, with wakefulness markedly higher than sleep (Figure 2A). Because the surrogate preserves each epoch’s waveform, this is *not* merely the changing non-sinusoidality of the dominant rhythm but a genuine change in $n:m$ coupling beyond the harmonic baseline; the all-moment entropy index carries it at least as strongly as the PLV. The same analysis between two distant derivations (Fpz–Cz versus Pz–Oz), against an IAAFT-of-one-channel null, shows *no* stage-dependent inter-regional coupling (Figure 2B). The honest reading: within-channel $n:m$ coupling — which at harmonically related frequencies partly reflects the rhythm’s waveform shape — carries a genuine, state-dependent component the new techniques track, whereas inter-regional $n:m$ coupling in scalp EEG sits at the noise floor, consistent with the broader difficulty of establishing cross-region $n:m$ coupling in macroscopic recordings. Both halves of this pattern replicate on an independent, higher-density recording with a different reference montage (four referential channels, Fpz–Cz’s bipolar derivation replaced by frontal/central/occipital electrodes): at matched power the within-channel stage-dependence recurs, while the between-channel effect is null across all six near- and far-electrode pairs — so the within-versus-between asymmetry is a property of the coupling, not of the two-channel sampling.

2.3 Four estimation choices decide whether a clean polyrhythm is recovered

On synthetic ground truth — a genuine 2:3 lock between two independent oscillators — an oracle (the $n:m$ PLV on band-pass–Hilbert phase at the known ratio) separates coupled from uncoupled cleanly ($\Delta = 0.79$). Four estimation choices decide whether an estimator reaches that separation: mishandling them drops the coupled–uncoupled margin to near zero ($\Delta = 0.02$; Figure 3A), and each is individually consequential. (i) *Convention*: the ratio rule returns multipliers under a

Stage-dependent $n:m$ phase coupling is genuine within-channel (delta) but absent between channels
 (entropy; Sleep-EDF; whole-spectrum, surrogate-controlled)
 A within-channel (Fpz-Cz) waveform-preserving (time-shuffle) null
 B between-channel (Fpz-Cz ↔ Pz-Oz) IAAFT null

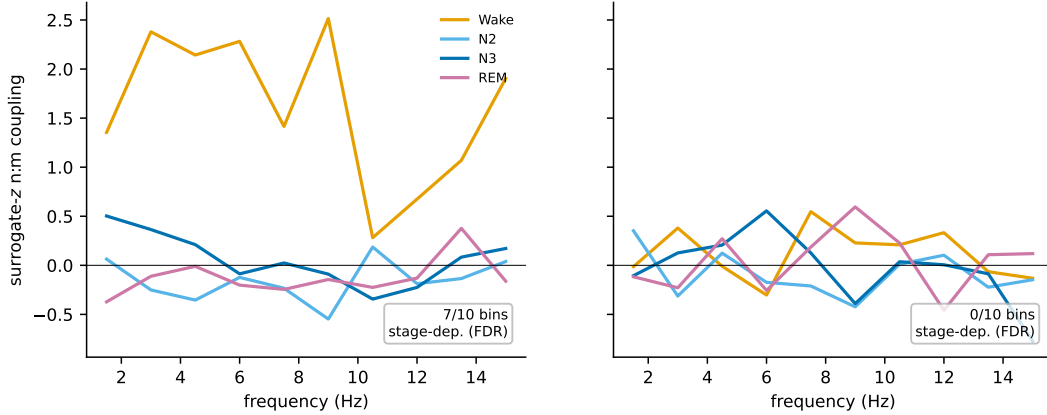


Figure 2: A state-dependent $n:m$ phase-coupling spectrum in sleep EEG. Surrogate- z $n:m$ phase coupling (entropy index) by sleep stage. (A) Within Fpz–Cz, against a waveform-preserving time-shuffle null: coupling is strongly stage-dependent in the delta band (wake $\gg$ sleep) and survives the waveform control. (B) Between Fpz–Cz and Pz–Oz, against an IAAFT null: no stage-dependent inter-regional coupling.

swapped convention applied literally, so the relative phase is non-stationary even for a perfect lock; the PLV then falls to $AUC = 0.38$ (0.17 against volume-conduction controls) and the zero-lag-rejecting metrics (PLI/wPLI/RRCi) drop to $AUC = 0.04\text{--}0.23$ on odd-integer ratios, below chance (Figure 3B). (ii) *Phase estimator*: the default short-time-Fourier-transform bin phase is leakage-limited and misses $n:m$ locking the Hilbert estimator recovers. (iii) *Ratio coverage*: the default ratio table reaches only 3:3 and silently relabels unmatched pairs as 1:1. (iv) *Weighting*: a complexity weight multiplied into the coupling value caps a perfect 2:3 lock far below its unweighted value. We correct each (canonical multipliers, Hilbert phase, exact-fraction ratios, unweighted readout).

2.4 The corrected estimator is sound across the ratio range

With the canonical convention, the $n:m$ PLV detects every tested lock from 1:2 through 5:7 at ceiling, identifies the true ratio against a grid of candidate multipliers with near-perfect accuracy, rises monotonically with coupling strength, and remains reliable to low SNR (Figure 4). The zero-lag-robust variants recover the lock at most ratios but plateau below ceiling; their robustness to instantaneous mixing buys little here because, at $n \neq m$, the two channels are read at different frequencies.

2.5 No single technique is best — the winner is regime-dependent

Across eight coupling regimes the techniques split into two families with complementary failure modes (Figure 5). On clean unimodal locks — and under non-sinusoidal coupling, non-stationary drift and added harmonics — the PLV is supreme ($AUC = 1.00$) and is the most sensitive at low SNR. But when the lock is *multimodal* (a multistable or antipodal relative phase) the PLV and the conditional-probability index *anti-detect* ($AUC \approx 0.34$), because a balanced multimodal distribution is *less* unimodally concentrated than the independent null. Here the all-moment indices dominate: the entropy index and phase mutual information recover the coupling ($AUC 0.90\text{--}0.99$) where every first-moment statistic fails. Weak graded coupling is genuinely hard for all techniques.

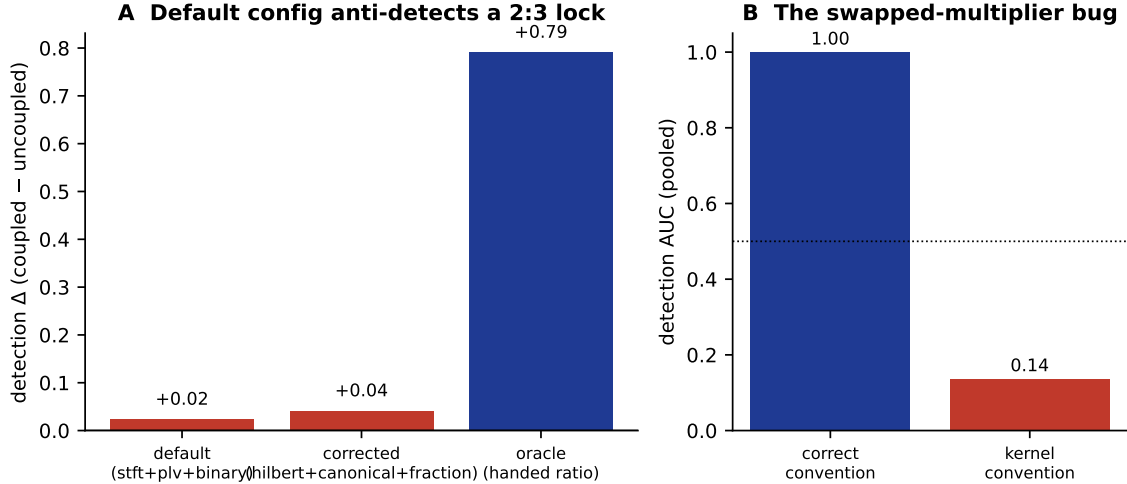


Figure 3: Four estimation choices decide recovery of a known lock. (A) Detection margin (coupled – uncoupled) for a clean 2:3 lock: the uncorrected configuration collapses to near zero ($\Delta = 0.02$), the corrected configuration improves it, and only the oracle separates cleanly ($\Delta = 0.79$). (B) The swapped-multiplier convention alone drives detection below chance; the canonical convention restores it.

The practical reading is a *panel*: the PLV for sensitivity on clean locks and at low SNR, the entropy/information indices for generality against multimodal locks. This pattern was reproduced independently, with separate generators and AUC code, under adversarial verification.

2.6 Within one signal, every technique false-positives — and no surrogate fully fixes it

A single non-sinusoidal oscillator has harmonics that are phase-locked by construction. Tested for “1:2 coupling” between its fundamental and second harmonic, *every* technique reports strong, significant coupling against an IAAFT null (surrogate z of 7, 42 and 37 for the PLV, entropy and mutual-information indices; all 100% significant; Figure 6A), although there is only one rhythm. The IAAFT null is the wrong control: it randomizes the cross-frequency phase relation, so a signal’s genuine harmonic lock appears “coupled” relative to the surrogate. Testing which null separates artifact from genuine coupling, no standard surrogate fully removes the artifact: a waveform-preserving block-shuffle reduces it roughly eight-fold while keeping genuine coupling strongly detectable, but a residual remains (Figure 6B). Within-signal $n:m$ at harmonically related frequencies is therefore fundamentally confounded with waveform shape. The sound regime is cross-signal: between independent sources the IAAFT-of-one-channel null destroys a real inter-signal relationship with no within-signal harmonic confound — the regime in which the ground-truth recovery above holds, and the regime our detector enforces.

2.7 A validated detector

We package these results as a cross-signal detector. It band-passes each component in its own band, applies the canonical multipliers, reports a panel of complementary techniques (the PLV plus the entropy and mutual-information indices), each with an IAAFT surrogate z and rank- p , scans candidate ratios for specificity, and emits an explicit scope-guard warning when the two inputs are too correlated to separate coupling from shared-source harmonics. On held-out ground truth

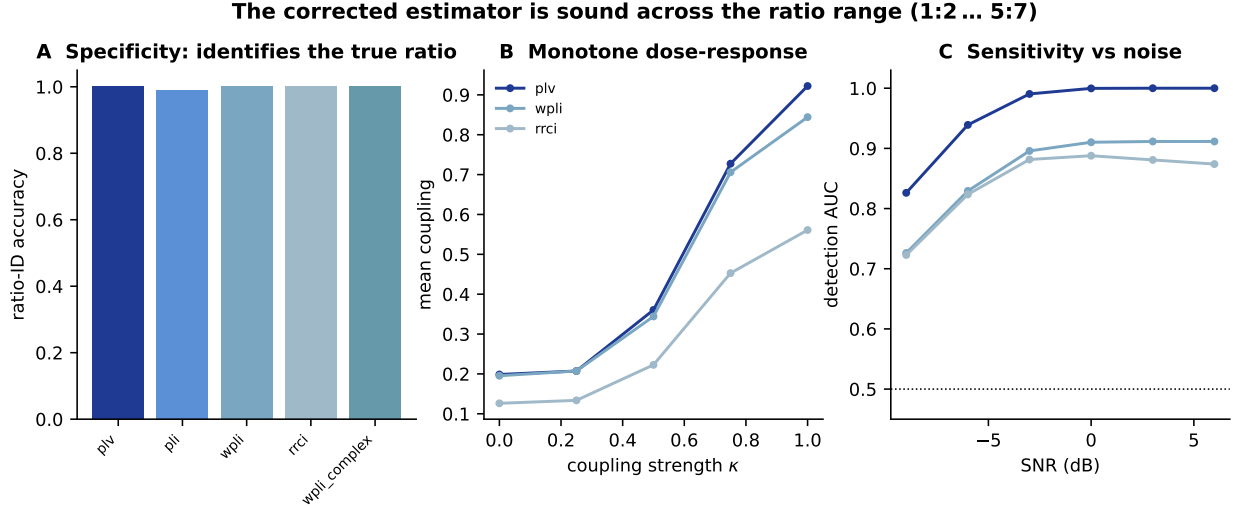


Figure 4: The corrected estimator is sound. (A) Ratio-identification accuracy across techniques. (B) Monotone dose-response in coupling strength κ . (C) Detection AUC versus SNR; the PLV reaches ceiling by 0 dB.

it recovers genuine locks across the ratio range, stays null for independent signals and off-target ratios, detects the multimodal locks the PLV alone misses, and warns on within-signal input. The detector and its regression tests are released in the framework.

3 Discussion

Treated as an experiment, $n:m$ coupling *measurement* yields three lessons. First, estimator defaults can be not merely weak but *directionally wrong*: a plausible pipeline recovered almost none of a textbook polyrhythm and, under a swapped ratio convention, scored genuine coupling below chance — and the failure was invisible without ground truth. Second, there is *no universal $n:m$ statistic*: the first-moment PLV family and the all-moment entropy/information family have complementary blind spots, so a panel — not a single default — is the honest instrument. Third, the *harmonics confound is a property of the design, not of the metric*: every technique false-positives within one non-sinusoidal signal and no surrogate fully removes it, so the only sound $n:m$ inference is between independent sources.

These results bear directly on the applications that motivate the measurement. A cardiorespiratory synchrogram used to grade anaesthetic depth, an autonomic coupling index in heart failure, or a cortico-muscular $n:m$ biomarker for deep-brain stimulation each rests on an estimator whose defaults, technique choice and surrogate we show to be consequential; a directionally wrong default or a harmonic false positive would corrupt the clinical readout. On the methodological critique that cross-frequency coupling claims are confounded by waveform shape and inflated by mismatched surrogates, the corrected estimator does not manufacture coupling where there is none, flags the within-signal regime where it cannot be trusted, and reports surrogate-calibrated inference rather than a raw statistic.

Limitations. The within-signal confound is scoped, not solved; a definitive within-signal test would require waveform-aware controls beyond phase surrogates. Weak graded coupling sits near the detection floor for all techniques. We did not implement a debiased estimator (pairwise phase

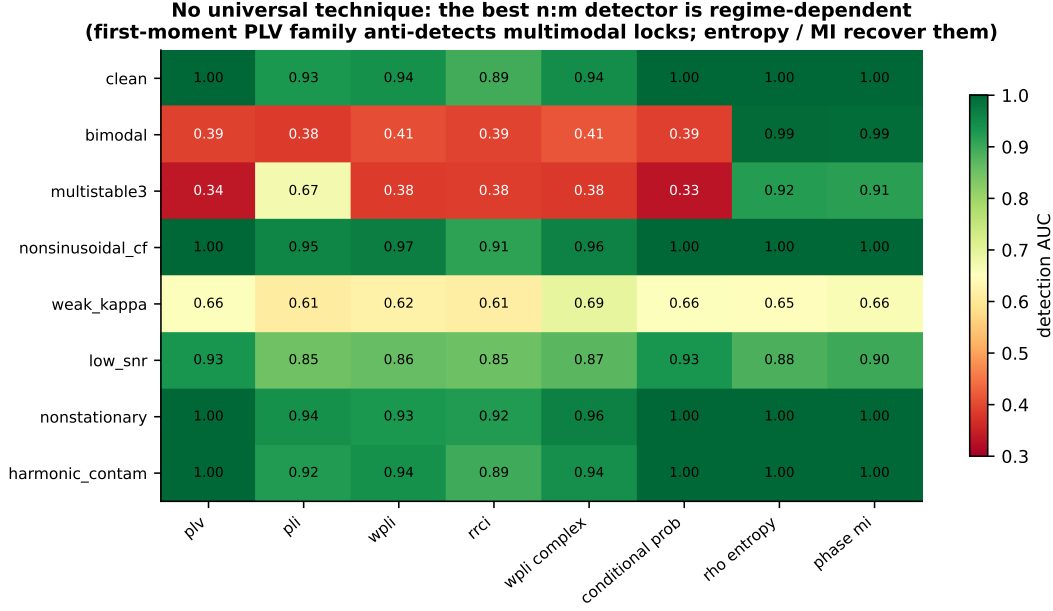


Figure 5: No universal technique. Detection AUC for eight techniques (columns) across eight coupling regimes (rows). The first-moment PLV family (left) is supreme on unimodal-style regimes but anti-detects the multimodal ones (*bimodal*, *multistable3*), where the all-moment entropy and mutual-information indices (right) dominate.

consistency, Vinck et al., 2010) for comparing coupling across epochs of differing length, a useful addition where trial counts vary. The real-data evidence spans a clearly-locked cardiorespiratory subject, a within-channel sleep effect replicated across two montages, and the established literature; a population study across full cohorts and conditions (paced breathing, sleep, exercise) would strengthen the prevalence estimate. Inter-regional scalp coupling is the hard case: in a separate music-listening EEG dataset, cross-regional $n:m$ coupling did not survive an across-subject control for the shared auditory stimulus, echoing the null between-channel sleep result — consistent evidence that genuine $n:m$ coupling between distant scalp regions is at or below the detection floor, and that within-signal positives must be read with the waveform caveat in mind.

References

- Aru J, Aru J, Priesemann V, Wibral M, Lana L, Pipa G, Singer W, Vicente R (2015). Untangling cross-frequency coupling in neuroscience. *Curr Opin Neurobiol* 31:51–61.
- Belluscio MA, Mizuseki K, Schmidt R, Diba K, Buzsáki G (2012). Cross-frequency phase–phase coupling between theta and gamma oscillations in the hippocampus. *J Neurosci* 32:423–435.
- Bramble DM, Carrier DR (1983). Running and breathing in mammals. *Science* 219:251–256.
- Cole SR, Voytek B (2017). Brain oscillations and the importance of waveform shape. *Trends Cogn Sci* 21:137–149.
- Palva JM, Palva S, Kaila K (2005). Phase synchrony among neuronal oscillations in the human cortex. *J Neurosci* 25:3962–3972.
- Purdon PL et al. (2013). Electroencephalogram signatures of loss and recovery of consciousness from propofol. *PNAS* 110:E1142–E1151.

$n:m$ at harmonically related frequencies is confounded within one signal → sound only cross-signal

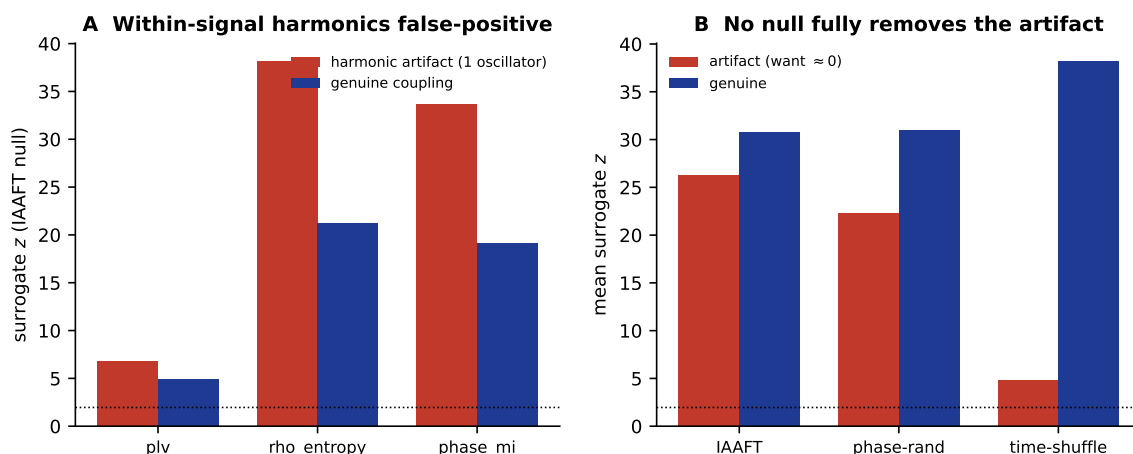


Figure 6: The harmonics confound. (A) Within a single non-sinusoidal oscillator, every technique reports significant “ $n:m$ coupling” under an IAAFT null — a false positive. (B) No standard null removes it: IAAFT and phase-randomization fail badly; a waveform-preserving time-shuffle reduces the artifact ~ 8 -fold but leaves a residual, while keeping genuine cross-signal coupling detectable.

Schäfer C, Rosenblum MG, Kurths J, Abel HH (1998). Heartbeat synchronised with ventilation. *Nature* 392:239–240.

Schäfer C, Rosenblum MG, Abel HH, Kurths J (1999). Synchronization in the human cardiorespiratory system. *Phys Rev E* 60:857–870.

Scheffer-Teixeira R, Tort ABL (2016). On cross-frequency phase–phase coupling between theta and gamma oscillations in the hippocampus. *eLife* 5:e20515.

Schnitzler A, Gross J (2005). Normal and pathological oscillatory communication in the brain. *Nat Rev Neurosci* 6:285–296.

Tass P, Rosenblum MG, Weule J, Kurths J, Pikovsky A, Volkmann J, Schnitzler A, Freund HJ (1998). Detection of $n:m$ phase locking from noisy data: application to magnetoencephalography. *Phys Rev Lett* 81:3291–3294.

van Leeuwen P et al. (2009). Influence of paced maternal breathing on fetal–maternal heart rate coordination. *PNAS* 106:13661–13666.

Vinck M, van Wingerden M, Womelsdorf T, Fries P, Pennartz CMA (2010). The pairwise phase consistency: a bias-free measure of rhythmic neuronal synchronization. *NeuroImage* 51:112–122.